Language acquisition follows conjugate Dirichlet updating I

Where the first study fused fixed beliefs in a single round, the second asks whether a single agent can *acquire* the likelihood of its shared world at all, and how quickly it does so. The design is a categorical source-mechanism analogue of the language-acquisition mechanism discussed by Friston et al. (Friston et al. 2024): each configured seed runs an agent that learns the likelihood of its shared world by conjugate Dirichlet updates over 24 count batches, the count update eq. 14. The recorded trajectory is $\mathrm{KL}(\text{true } A \,\|\, \text{learned } A)$ before each batch — it starts at the flat-prior maximum and declines monotonically toward zero as the agent "acquires the language" of its world. The KL here is the same divergence whose $\alpha \to 1$ Rényi limit is established by Lemma 3 (eq. 3), so the learning curve and the recovery limits measure the same object.

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- Initial KL (flat prior): 3.4231
- Final KL (after 24 batches): 0.0027
- Total KL reduction: 3.4204
- Trajectory points: 25 ordered learning steps per seed
- Pointwise seed bootstrap: 95% intervals over $n = 480$ independent seeds
- Trajectory monotone-decreasing: Yes

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The monotone decline to a final KL of 0.0027 is the demonstrated quantity behind the acquisition claim: under the tested count schedule, the learned likelihood moves toward the true generative likelihood as conjugate counts accumulate. This finite trajectory is evidence of the update's behavior, not a convergence-rate result for arbitrary data-generating processes.

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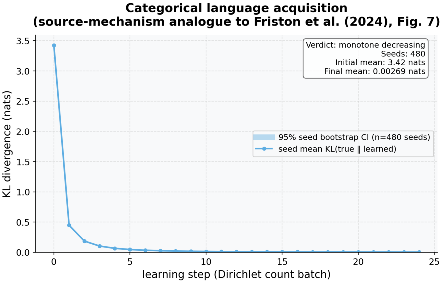

Figure 1: Seed-mean KL divergence from the true likelihood A to the learned likelihood A. The plotted quantity is $\mathrm{KL}(\text{true } A \,\|\, \text{learned } A)$. Source relation: source-mechanism analogue to Friston et al. (2024), Fig. 7; estimand: seed-mean KL in nats by ordered learning step. x-axis: ordered Dirichlet count batch, from the flat prior at zero through all 24 batches (25 points per seed); y-axis: summed per-column KL divergence between the true likelihood and the current expected likelihood, in nats. The solid line is the mean over 480 independent configured seeds, and the shaded band is the pointwise 95% percentile-bootstrap interval resampling seeds at each learning step. The replication unit is seed, not the ordered trajectory points. The mean curve falls monotonically from 3.4231 nats to 0.0027 nats (total reduction 3.4204 nats, computed from the unrounded endpoints); the computed monotone-decreasing verdict is Yes. This reduced categorical protocol is related to, but does not exactly reproduce, the richer multi-episode protocol in Friston et al. (2024).

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fig. 1 plots the full learning curve and its CI band.